

09/16/2026: Sets

CSCI 246: Discrete Structures

Textbook reference: Sec. 10, Scheinerman

Graded Quiz Pickup

Quizzes are in the front of the room, grouped into four bins (A-G, H-L, M-R, S-Z) by last name. The quizzes are upside down with your last name on the back. Come find yours before, during, or after class. Only turn the quiz over if it's yours.

Early Feedback

1. What do like about class so far / what's helping with your learning?
2. What do dislike about class so far / what's interfering with your learning?
3. Any other comments?

Today's Agenda

- Review group exercises ($\approx$ 15 mins)
- Mini-lecture ($\approx$ 15 mins)
- Group exercises ($\approx$ 15 mins)

Study Guide For Quiz 3

Readings

Sec. 7 (Boolean Algebra)

- Know how to prove Props 7.1 and 7.3 using truth tables.
- Understand the Boolean algebra properties listed in Theorem 7.2. (What are they saying? Could you use them?)

Sec. 8 (Lists)

- Know how to solve examples 8.1, 8.3, 8.4, and 8.5
- Know how to apply Theorems 8.2 and 8.6.

Sec. 9 (Factorial)

- Know how to apply the definition of factorial.
- Know how to justify why $0! = 1$.
- Understand how to read product notation.

Group exercises

Know how to do all of the group exercises from those course meetings.

Review group exercises

Some notes on Sets I (Introduction, Subsets)

Subtopics from reading

Scheinerman Sec. 10 (Sets I: Introduction, Subsets) covers the following subtopics

1. Introduction to sets and set notation,
2. How to show one set is a subset of the other,
3. How to show two sets are equal, and
4. How to count the number of subsets of a set.

Today's mini-lecture and group exercises focus on these subtopics.

Outline for Mini-Lecture:

- ● Introduction to sets and set notation
- ● How to show two sets are equal
- ● How to count the number of subsets

Sets vs. lists

Question: Where did Mike go on vacation last summer?

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Answer via set: Sets are **unordered**. I could equivalently provide any of the following as a solution.

$\{\text{Florida, camping, lake}\} = \{\text{Florida, lake, camping}\}$
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Three special sets of numbers

Definition

The **integers** are the positive whole numbers, the negative whole numbers, and zero. That is, the set of integers, denoted by $\mathbb{Z}$ is

$$\mathbb{Z} = \{\dots, -3, -2, -1, 0, 1, 2, 3\dots\}.$$

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The **rational numbers** (denoted $\mathbb{Q}$) are the numbers formed by dividing two integers a/b , where $b \neq 0$. That is,

$$\mathbb{Q} = \{a/b : a, b \in \mathbb{Z}, b \neq 0\}$$

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**Set builder
notation**

Outline for Mini-Lecture:

- ● Introduction to sets and set notation
- ● How to show two sets are equal
- ● How to count the number of subsets

Applications in computer science: Some examples

In computer science, proving set equality shows up in:

- **Database theory:** A query optimizer may rewrite `WHERE id IN (SELECT ...)` as a `JOIN`. That rewrite is legal only if both queries return the *same set* of rows.

Applications in computer science: Some examples

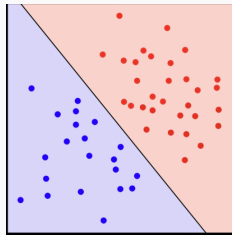
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- **Formal languages.** $(a|b)^*$ and $(a^*b^*)^*$ look different but describe the same language: all strings over $\{a, b\}$. Regex engines rely on such equivalences to simplify patterns.

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- **Neural networks:** A single-layer perceptron represents *exactly* the linearly separable classifiers. **XOR is not in that set**, which is why neural nets need hidden layers.

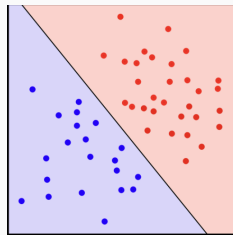


Linearly separable: one straight line splits the two classes.

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- **Neural networks:** A single-layer perceptron represents *exactly* the linearly separable classifiers. **XOR is not in that set**, which is why neural nets need hidden layers.
- **Complexity theory.** If $P = NP$, thousands of problems – scheduling, vehicle routing, compiler register allocation — would all get polynomial-time algorithms.



Linearly separable: one straight line splits the two classes.

Problem

Prove that the following two sets are equal:

$$E = \{x \in \mathbb{Z} : x \text{ is even}\}, \text{ and}$$

$$F = \{x \in \mathbb{Z} : x = a + b, \text{ where } a \text{ and } b \text{ are both odd}\}$$

Puzzle

Evaluate the proposed solution below (provided by a student last semester).

Proposed Solution

If a and b are both odd, then we can write $a = 2c + 1$ and $b = 2d + 1$, where both c and d are integers. Now

$$a + b = (2c + 1) + (2d + 1) = 2c + 2d + 2 = 2(c + d + 1).$$

That is, $a + b = 2e$, where $e := c + d + 1$ is an integer. Hence, $a + b$ is even. Hence, the sets E and F are equal.

How to show two sets are equal

Proving two sets are equal

To show $A = B$, we show $A \subseteq B$ and $B \subseteq A$.

- $A \subseteq B$. We show that if $x \in A$, then $x \in B$.
- $B \subseteq A$. We show that if $x \in B$, then $x \in A$.

How to show two sets are equal

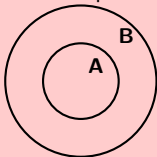
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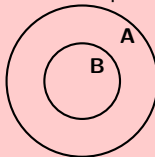
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Visualization

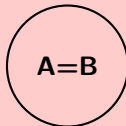
The first bulletpoint shows



The second bulletpoint shows



Therefore



How to show two sets are equal: Some intuition

Suppose you're **packing for a trip**. Your bag matches your checklist exactly when

1. $\text{checklist} \subseteq \text{bag}$: nothing is **missing**,
2. $\text{bag} \subseteq \text{checklist}$: nothing **extra** snuck in.



Solution to problem

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Solution. To show $E = F$, we show $E \subseteq F$ and $F \subseteq E$.

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Solution. To show $E = F$, we show $E \subseteq F$ and $F \subseteq E$.

- $F \subseteq E$. We show that if $x \in F$, then $x \in E$. Let $x \in F$. Hence $x = a + b$ for odd integers a, b . By the definition of odd, we can write $a = (2c + 1)$ and $b = (2d + 1)$ for integers c, d . Hence

$$x = a + b = (2c + 1) + (2d + 1) = 2c + 2d + 2 = 2(c + d + 1)$$

That is, $a + b = 2e$, where $e := c + d + 1$. Hence, $a + b$ is even. Hence, $x \in E$.

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That is, $a + b = 2e$, where $e := c + d + 1$. Hence, $a + b$ is even. Hence, $x \in E$.

- $E \subseteq F$. We show that if $x \in E$, then $x \in F$. Let $x \in E$. Then $x = 2a$ for some integer a . Now we write

$$x = 2a = \underbrace{(2a + 1)}_{\text{odd (by def.)}} + \underbrace{(-1)}_{\text{odd}}$$

Hence x is the sum of two odd numbers. Hence $x \in F$.

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Counting Subsets

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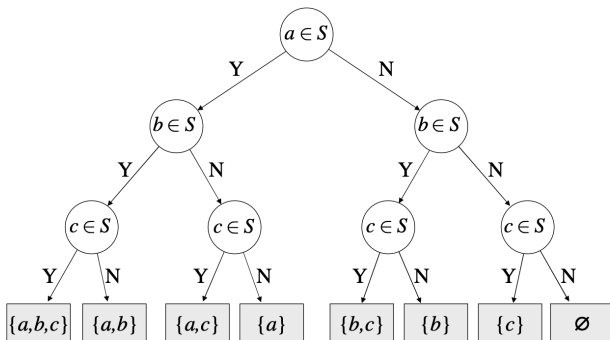


FIGURE 1.2. A decision tree for selecting a subset of $\{a, b, c\}$.

Counting Subsets

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Explanation. We can form a correspondence between lists and subsets since each subset is formed by a yes vs. no decision on whether to include each set item. For example, if $A = \{a, b, c\}$, we have

List		Subset
(Y,Y,Y)	$\leftrightarrow$	$\{a, b, c\}$
(Y,Y,N)	$\leftrightarrow$	$\{a, b\}$
(Y,N,Y)	$\leftrightarrow$	$\{a, c\}$
(Y,N,N)	$\leftrightarrow$	$\{a\}$
(N,Y,Y)	$\leftrightarrow$	$\{b, c\}$
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(N,Y,Y)	$\leftrightarrow$	$\{b, c\}$
(N,Y,N)	$\leftrightarrow$	$\{b\}$
(N,N,Y)	$\leftrightarrow$	$\{c\}$
(N,N,N)	$\leftrightarrow$	$\emptyset$

Recall Scheinerman Theorem 8.6: the number of lists of length k whose elements are chosen from a pool of n elements is n^k if repetitions are permitted. For counting subsets, $n = 2$ (yes vs. no) and $k = |A|$. In this case, we can form $2^{|A|} = 2^3 = 8$ such lists.

Counting Subsets: Theorem

Below we provide two versions of the same theorem.

Lovasz et al. Theorem 1.3.1

A set with n elements has 2^n subsets.

Scheinerman Theorem 10.7

Let A be a finite set. The number of subsets of A is $2^{|A|}$.

Enumerating Subsets

0	$\Leftrightarrow$	0_2	$\Leftrightarrow$	000	$\Leftrightarrow$	$\emptyset$
1	$\Leftrightarrow$	1_2	$\Leftrightarrow$	001	$\Leftrightarrow$	$\{c\}$
2	$\Leftrightarrow$	10_2	$\Leftrightarrow$	010	$\Leftrightarrow$	$\{b\}$
3	$\Leftrightarrow$	11_2	$\Leftrightarrow$	011	$\Leftrightarrow$	$\{b, c\}$
4	$\Leftrightarrow$	100_2	$\Leftrightarrow$	100	$\Leftrightarrow$	$\{a\}$
5	$\Leftrightarrow$	101_2	$\Leftrightarrow$	101	$\Leftrightarrow$	$\{a, c\}$
6	$\Leftrightarrow$	110_2	$\Leftrightarrow$	110	$\Leftrightarrow$	$\{a, b\}$
7	$\Leftrightarrow$	111_2	$\Leftrightarrow$	111	$\Leftrightarrow$	$\{a, b, c\}$

Source: Lovasz, Pelikan, Vesztergombi. *Discrete Mathematics: Elementary and Beyond*.

Remark. Using a code like this, you could determine that the 233rd subset of a 10-element set $\{a_1, \dots, a_{10}\}$ consists of elements $\{a_3, a_4, a_5, a_7, a_{10}\}$. (That's because 233 in binary is 11101001, which corresponds to the code 0011101001.)

Feature selection in machine learning. If you have n possible features, there are 2^n possible subsets of features you could choose. Example: With 20 features, there are $2^{20} \approx 1$ million possible subsets.

Random group assignments

Ashley, Isabelle: 12

Bailey, Tanna: 10

Blaisdell, Olyver: 12

Brown, Ashton: 13

Brown, Jonathan: 1

Cardenas, Dominic: 9

Cutler, George: 13

Fabik, Roland: 6

Fellows, Audrey: 9

Fitzgerald, Ryan: 7

Fuller, Ryan: 11

Griffen, Clara: 3

Grutsch, Tanner: 4

Hager, Nathan: 6

Horner, Jayden: 10

Johnson, Jaret: 7

Kintzel, Samantha: 4

Lameres, Alexis: 10

Lee, Holden: 7

Mayala, Reece: 8

McLean, Connor: 2

Mcbride, Daniel: 2

Miller, Maverick: 11

Moss, Emily: 9

Mousseau, Matthew: 5

Neuman, Evan: 8

Putnam, John: 5

Ramos, Vydel: 3

Rathbun, Jedidiah: 8

Roberts, Matthew: 11

Roller, Nathan: 3

Ryan, Otis: 2

Shevtsov, Timothy: 5

Sinclair, Aidan: 12

Smith, Charles: 1

Stensland, Emma: 1

Thiessen, Logan: 6

Thompson, Jaiden: 13

Vath, Ethan: 4

Group exercises: Sets

- Find the cardinality of the following sets
 - $\{x \in \mathbb{Z} : |x| \leq 10\}$
 - $\{x \in \mathbb{Z} : 1 \leq x^2 \leq 2\}$
 - $\{x \in \mathbb{Z} : x \in \emptyset\}$
 - $2^{\{1,2,3\}}$
 - $\{x \in 2^{\{1,2,3,4\}} : |x| = 1\}$
 - $\{\{1,2\}, \{3,4,5\}\}$
- Let $A = \{x \in \mathbb{Z} : 4|x\}$ and $B = \{x \in \mathbb{Z} : 2|x\}$. Prove that $A \subseteq B$.
- Let $A = \{0,1,2,3,4\}$ and $B = \{x \in \mathbb{N} : x^2 < 17\}$. Prove that $A = B$.

Solution to group exercise #1 (a-d)

Solution.

$$\text{a) } \left| \{x \in \mathbb{Z} : |x| \leq 10\} \right| = \left| \{-10, \dots, -1, 0, 1, \dots, 10\} \right| = 21$$

Solution to group exercise #1 (a-d)

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$$\text{c) } \left| \{x \in \mathbb{Z} : x \in \emptyset\} \right| = \left| \emptyset \right| = 0$$

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$$\text{c) } \left| \{x \in \mathbb{Z} : x \in \emptyset\} \right| = \left| \emptyset \right| = 0$$

d) To find $\left| 2^{2^{\{1,2,3\}}} \right|$, first recall from the text that for any finite set A , 2^A refers to the set of all subsets of A . So setting $A := \{1, 2, 3\}$, we have

$$B := 2^A = 2^{\{1,2,3\}} = \left\{ \emptyset, \{1\}, \{2\}, \{3\}, \{1, 2\}, \{1, 3\}, \{2, 3\}, \{1, 2, 3\} \right\}.$$

Now 2^B is the set of all subsets of *that*, so

$$2^B = \left\{ \emptyset, \{\{1\}\}, \{\{2\}\}, \{\{3\}\}, \{\{1\}, \{2\}\}, \dots, \{\{1\}, \{1, 2, 3\}, \{2\}, \{1, 2\}\}, \dots \right\}$$

Notably, $B = 2^A$ has $2^{|A|} = 2^3 = 8$ elements, and so 2^B has $2^{|B|} = 2^8$ elements. Hence, $\left| 2^{2^{\{1,2,3\}}} \right| = 2^8$.

Solution to group exercise #1 (e-f)

Solution.

e) To find $\left| \{x \in 2^{\{1,2,3,4\}} : |x| = 1\} \right|$, we proceed similarly as in part (d). The set $B := 2^{\{1,2,3,4\}}$ refers to the set of all subsets of $\{1, 2, 3, 4\}$. That is,

$$B = \left\{ \emptyset, \{1\}, \{2\}, \{3\}, \{4\}, \{1, 2\}, \{1, 3\}, \{1, 4\}, \{2, 4\}, \dots, \{1, 2, 3, 4\} \right\}.$$

In particular B has $2^{|\{1,2,3,4\}|} = 2^4 = 16$ elements. Only four of those elements have cardinality 1, namely $\{1\}$, $\{2\}$, $\{3\}$, and $\{4\}$. Hence

$$\left| \{x \in 2^{\{1,2,3,4\}} : |x| = 1\} \right| = 4.$$

Solution to group exercise #1 (e-f)

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$$\left| \{x \in 2^{\{1,2,3,4\}} : |x| = 1\} \right| = 4.$$

$$\text{f) } \left| \{\{1, 2\}, \{3, 4, 5\}\} \right| = 2$$

Solution to group exercise #2

Problem. Let $A = \{x \in \mathbb{Z} : 4|x\}$ and $B = \{x \in \mathbb{Z} : 2|x\}$. Prove that $A \subseteq B$.

Solution.

Annotation	Main Text
Convert Prop. to "if-then" form	We show that if $x \in A$, then $x \in B$.
State assumption ("if")	Let $x \in A$.
Unravel defs.	Since, $x \in A$, we know that $4 x$. So by the definition of divisibility, there is some $n \in \mathbb{Z}$ such that $x = 4n$.
*** The glue ***	We write $x = 4n = 2(2n)$
Unravel defs.	So there is an integer $c = 2n$ such that $x = 2c$. So $2 x$.
State conclusion	Hence, $x \in B$.

Solution to group exercise #3 (Big picture)

Problem. Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{x \in \mathbb{N} : x^2 < 17\}$. Prove that $A = B$.

Structure of solution. To show $A = B$, we show $A \subseteq B$ and $B \subseteq A$.

- $A \subseteq B$. We show that if $x \in A$, then $x \in B$.
- $B \subseteq A$. We show that if $x \in B$, then $x \in A$.

Solution to group exercise #3

Problem. Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{x \in \mathbb{N} : x^2 < 17\}$. Prove that $A = B$.

Solution. To show $A = B$, we show $A \subseteq B$ and $B \subseteq A$.

Solution to group exercise #3

Problem. Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{x \in \mathbb{N} : x^2 < 17\}$. Prove that $A = B$.

Solution. To show $A = B$, we show $A \subseteq B$ and $B \subseteq A$.

- $A \subseteq B$. We show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $x \in \{0, 1, 2, 3, 4\}$. Then $x^2 \in \{0, 1, 4, 9, 16\}$. So $x^2 < 17$. Hence $x \in B$.

Solution to group exercise #3

Problem. Let $A = \{0, 1, 2, 3, 4\}$ and $B = \{x \in \mathbb{N} : x^2 < 17\}$. Prove that $A = B$.

Solution. To show $A = B$, we show $A \subseteq B$ and $B \subseteq A$.

- $A \subseteq B$. We show that if $x \in A$, then $x \in B$. Let $x \in A$. Then $x \in \{0, 1, 2, 3, 4\}$. Then $x^2 \in \{0, 1, 4, 9, 16\}$. So $x^2 < 17$. Hence $x \in B$.
- $B \subseteq A$. We show that if $x \in B$, then $x \in A$. Let $x \in B$. Hence $x \in \mathbb{N}$ and $x^2 < 17$. By taking the square root of each side, $x^2 < 17$ implies $|x| < \sqrt{17}$. The only natural numbers satisfying $|x| < \sqrt{17}$ are 0, 1, 2, 3 and 4. Hence $x \in A$.